MCS Saturday 7th grade SHSAT Math

Grade 7 SHSAT Math · New CAT Digital Format

Ratio & Proportion Control Lab

Session 5 Homework Activity Sheet · All-New Questions

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https://hue881.github.io/mcs/2026/fall/math/7th-grade/

Name _____ **Date** _____

Completion Check	___ /18	___ /8	___ /26
Representation + accuracy + CAT commit routine	Core	Transfer	Total

A CAT Relationship Protocol

5 prompts · 7 min

Rehearse the decision sequence you will use on a forward-only screen. State observable actions rather than “be careful.”

1 · Label | **2 · Model** | **3 · Scale** | **4 · Verify** | **5 · Commit**

A1 · Ratio order

Recall

Before calculating a ratio, what two labels will you write so that you do not reverse its order?

State the label routine precisely.

Write the first named quantity, then the second;
label the ratio first quantity : second quantity.

A2 · Whole check

Explain

A question asks for **cats to all pets**. Explain the denominator check that distinguishes part-to-whole from part-to-part.

Complete: The denominator must represent ...

all pets, including cats and every other pet,
not just the pets that are not cats.

A3 · Structure scan

Apply

A table includes the pairs **(4, 18)** and **(10, 45)**. What one quotient should you compare before deciding that the relationship is proportional?

Name the quotient with labels y/x (output/input)

Do not calculate both values; name the structural check.

A4 · Reasonableness

Reason

Your model says a group representing 3 of 8 equal parts is larger than the whole. Identify the contradiction and your first audit.

Contradiction + first place to check

3/8 < 1, so the part must be smaller.
First check for an inverted fraction (8/3).

A5 · Final commit

Defend

Write a two-sentence commit rule for a multi-step ratio item: one sentence for checking the unchanged relationship and one for checking the requested quantity.

Sentence 1: relationship check. Sentence 2: target and format check.

I verify that each unchanged ratio or unit rate still matches the given relationship.
I check that my answer is the requested quantity and uses the required units and entry format.

B Representation Warm-Up

6 questions · 10 min

Label each quantity before calculating. Show a total-parts model, scale arrow, unit rate, or constant of proportionality.

B1

Level 1

Write the ratio of **15 triangles** to **25 circles** in simplest form.

Labels + common factor

Triangles : circles = 15 : 25
Divide both counts by 5: 3 : 5.

Typed entry 3 : 5

B2

Level 1

In an ensemble, 6 students play violin and 18 play other instruments. Find the ratio of violin players to **all** students.

Build the whole first

All students = 6 + 18 = 24.
Violin : all = 6 : 24 = 1 : 4.

Typed entry 1 : 4

B3

Level 2

A snack mix uses 4 cups of pretzels for every 11 cups of crackers. How many cups of pretzels pair with 33 cups of crackers?

Show one scale arrow

Crackers: 11 x 3 = 33 cups.
Pretzels: 4 x 3 = 12 cups.

Typed entry 12

B4

Level 2

Which pair is proportional to **16 : 28**?

A 8 : 13

B 20 : 35

C 24 : 40

D 32 : 54

Show the common multiplier or simplified ratios

B. 20 : 35
16 : 28 = 4 : 7; 20 : 35 = 4 : 7.
Both original entries are multiplied by 5/4.

B5

Level 2

The table shows direct variation. Find the missing value.

Find k = y/x

Cannot determine: the table is missing.
A known (x, y) pair is needed to find k;
then use y = kx for the missing value.

Typed entry Insufficient information

B6

Level 3

A jar has purple and silver beads in a ratio of **7 : 5**. Without finding either count, what fraction of all beads are silver? Explain the denominator.

Total parts + requested part

Silver / all = 5 / (7 + 5) = 5/12.
The denominator counts all 12 equal parts,
including both purple and silver beads.

Typed entry 5/12

C Core Accuracy Set

12 questions · 24 min

Treat each card as a separate CAT screen. Represent first, enter one response, perform one final check, and commit.

Label ratio order | Count total parts | Scale both quantities | Check the requested target

C1

Level 1

Simplify **42 : 63**.

Show your work.

42 / 21 = 2; 63 / 21 = 3.
Greatest common factor = 21.

Use a colon: 2 : 3

☐ I divided both entries by the same greatest common factor.

C2

Level 1

A basket holds 14 red apples and 21 green apples. What is the ratio of red apples to all apples?

Show your work.

All apples = 14 + 21 = 35.
Red : all = 14 : 35 = 2 : 5.

Simplified ratio: 2 : 5

☐ I included both colors in the whole.

C3

Level 2

Complete the proportion: **5/9 = n/54**.

Show your work.

54 / 9 = 6.
n = 5 x 6 = 30.

One integer: 30

☐ I used one multiplier on numerator and denominator.

C4

Level 2

The ratio of tulips to daisies is **5 : 8**. There are 156 flowers. How many are tulips?

Show your work.

Total parts = 5 + 8 = 13.
156 / 13 = 12 flowers per part.
Tulips = 5 x 12 = 60.

One integer: 60

☐ My 13 ratio parts rebuild the total.

C5

Level 2

A cyclist travels 42 miles in 3.5 hours at a constant speed. Find the unit rate in miles per hour.

Show your work.

42 miles / 3.5 hours = 12 miles/hour.

Number only: 12

☐ My quotient is miles divided by hours.

C6

Level 3

y varies directly with **x**. If **y = 31.5** when **x = 14**, find **y** when **x = 22**.

Show your work.

k = 31.5 / 14 = 2.25.
y = 2.25 x 22 = 49.5.

One value: 49.5

☐ I kept y/x constant.

C Core Accuracy Set · continued

12 questions · 24 min

Treat each card as a separate CAT screen. Represent first, enter one response, perform one final check, and commit.

Label ratio order | Count total parts | Scale both quantities | Check the requested target

C7 · TEI

Level 3

Select all equations that represent direct variation with constant **k = 2.4**.

A y = 2.4x

B y/x = 2.4

C x/y = 2.4

D y = x + 2.4

Show your work.

A and B: both state y = 2.4x.
C has k = 1/2.4; D has a nonzero intercept.

☐ Each selected form has the same multiplier.

C8

Level 3

The angles of a triangle are in the ratio **3 : 5 : 7**. Find the largest angle.

Show your work.

3 + 5 + 7 = 15 parts; 180 / 15 = 12.
Largest = 7 x 12 = 84 degrees.
Check: 36 + 60 + 84 = 180.

No degree symbol 84

☐ My three angle measures sum to 180.

C9

Level 3

On a trail map, 4.2 centimeters represents 15 kilometers. A route measures 9.8 centimeters. Find its actual length.

Show your work.

Actual length = 9.8 x (15 / 4.2)
= 35 kilometers.

Number only 35

☐ Drawing units and actual units stayed aligned.

C10

Level 3

A 192-gram alloy contains copper and zinc in a ratio of **7 : 9**. How many grams are copper?

Show your work.

Total parts = 7 + 9 = 16.
192 / 16 = 12 grams per part.
Copper = 7 x 12 = 84 grams.

One integer 84

☐ I used 16 total parts and selected 7.

C11

Level 4

A photo is enlarged from 8 inches wide to 14 inches wide. Its original height is 10 inches. Find the new height if the dimensions remain proportional.

Show your work.

Scale factor = 14 / 8 = 1.75.
New height = 10 x 1.75 = 17.5 inches.

Number only 17.5

☐ Width and height used the same scale factor.

C12

Level 4

Machine A packs 180 boxes in 12 minutes. Machine B packs 272 boxes in 16 minutes. Which machine has the greater constant rate, and by how many boxes per minute?

Find and label both unit rates

A: 180/12 = 15 boxes/min.
B: 272/16 = 17 boxes/min; 17 - 15 = 2.

Final response B, by 2 boxes/min

☐ I compared rates with the same units.

D SHSAT Transfer Set

8 questions · 22 min

Separate initial and changed states. Preserve only relationships the story says remain constant, then answer the requested quantity.

D1 · Changed state

Level 3

At a school play, adult tickets to student tickets were sold in a ratio of **4 : 9**, for 351 tickets total. Then 27 student tickets were refunded. What is the new ratio of adult tickets to student tickets?

Original counts → changed count → simplify

Total parts = 4 + 9 = 13.

Each part = 351 / 13 = 27 tickets.

Adults = 4 x 27 = 108.

Students originally = 9 x 27 = 243.

Students after refunds = 243 - 27 = 216.

New adult : student ratio = 108 : 216 = 1 : 2.

New ratio 1 : 2

D2 · Three-part ratio

Level 3

Green, yellow, and white beads are in a ratio of **3 : 4 : 5**, with 288 beads total. After 24 yellow beads are removed and 12 green beads are added, find the ratio of green beads to all remaining beads.

Rebuild each group and the new total

Total parts = 3 + 4 + 5 = 12.

Each part = 288 / 12 = 24 beads.

Original: green 72, yellow 96, white 120.

New: green 84, yellow 72, white 120.

New total = 84 + 72 + 120 = 276.

Green : all = 84 : 276 = 7 : 23.

Simplified ratio 7 : 23

D3 · Scale recipe

Level 3

A bakery uses 2.25 cups of flour for 18 muffins. At the same rate, how many cups are needed for 50 muffins?

Unit rate or equivalent ratio

Flour per muffin = 2.25 / 18 = 0.125 cup.

For 50 muffins: 50 x 0.125 = 6.25 cups.

Check: 6.25 / 50 = 2.25 / 18.

Cups 6.25

D4 · Similar figures

Level 3

Two similar rectangles have length ratio **5 : 8**. The smaller rectangle has width 7.5 centimeters. Find the larger width.

Match corresponding dimensions

Larger / smaller scale factor = 8 / 5.

Larger width = 7.5 x (8 / 5) = 12 cm.

Check: 7.5 / 12 = 5 / 8.

Centimeters 12

Separate initial and changed states. Preserve only relationships the story says remain constant, then answer the requested quantity.

D5 · Reverse total

Level 4

The ratio of fiction to nonfiction books on a cart is **11 : 6**. After 15 fiction books are removed, fiction books are exactly twice the nonfiction books. How many books were on the cart originally?

Model original groups with one variable

Let fiction = 11t and nonfiction = 6t.

After removal: 11t - 15 = 2(6t).

11t - 15 = 12t, so t = -15.

Negative book counts are impossible.

Also, 11/6 < 2; removing fiction only

decreases the fiction/nonfiction ratio.

No valid original total exists as written.

Original total No valid solution

D6 · Work rate

Level 4

Nine identical pumps fill 6 tanks in 14 hours. At the same combined rate per pump, how many hours will 12 pumps need to fill 8 tanks?

Compare tank-hours per pump or total pump-hours

9 x 14 = 126 pump-hours fill 6 tanks.

126 / 6 = 21 pump-hours per tank.

8 tanks need 8 x 21 = 168 pump-hours.

With 12 pumps: 168 / 12 = 14 hours.

Check: pumps and tanks both scale by 4/3,

so the time remains unchanged.

Hours 14

D7 · TEI

Select all

Select all data sets in which **y** varies directly with **x**.

- **A** (3, 8.1), (7, 18.9)
- **B** (4, 13), (8, 25)
- **C** (5, 2), (20, 8)
- **D** (6, 15), (10, 24)

Record the constant quotient for each selection

A and C

A: 8.1 / 3 = 18.9 / 7 = 2.7.

C: 2 / 5 = 8 / 20 = 0.4.

B: 13 / 4 = 3.25, but 25 / 8 = 3.125.

D: 15 / 6 = 2.5, but 24 / 10 = 2.4.

Only A and C have constant y/x.

D8 · Error analysis

Level 4

A student solves **18/30 = n/75** by adding 45 to the denominator and then adding 45 to the numerator, obtaining **n = 63**. Identify the exact structural error, find the correct value, and give a fast reasonableness check.

Error → correct multiplicative relationship → check

Adding the same amount does not preserve

a ratio; multiply both entries by one factor.

75 / 30 = 2.5, so n = 18 x 2.5 = 45.

Check: 18/30 = 3/5 = 0.6.

0.6 x 75 = 45, whereas 63/75 = 0.84.

Correct n 45

E One-Screen Rehearsal

3 decisions · 5 min

Assume you cannot return after submitting. Describe the final check you would perform in each situation.

E1. You found a scale factor of **3/5**, but the target quantity should be larger than its paired original quantity. What does that mismatch tell you to inspect?

Inspect the scale-factor direction and corresponding labels: use target / original.

A factor of 3/5 shrinks; if those quantities were reversed, the enlargement factor is 5/3.

E2. Your work ends with the constant of proportionality, but the prompt asks for an output at **x = 36**. State the missing action.

Substitute x = 36 into y = kx and calculate y = 36k.

Enter that output, not the constant k.

E3. Your calculation is correct, but the entry field requests a simplified ratio using a colon. State exactly what should—and should not—appear in the field.

Enter only the two whole-number terms in simplest form, in the requested order, separated by a colon.

Do not include labels, units, an equals sign, a fraction slash, or an explanation.

F Process Debrief

Evidence, not impressions

Complete this before checking any solutions. Focus on actions that can be repeated under forward-only CAT conditions.

- ☐ I labeled the first and second quantities before writing each ratio.
- ☐ I built the whole before answering a part-to-whole question.
- ☐ I multiplied both entries by the same factor.
- ☐ I used **y/x** consistently when finding a direct-variation constant.
- ☐ I separated original and changed states in multi-step stories.
- ☐ I checked that reconstructed parts returned to the given total.
- ☐ I answered the requested group—not merely one part or a scale factor.
- ☐ I entered only the requested value, ratio form, or unit-free response.

Timing record Pace audit

Record your time by section. Circle or describe the point where your pace changed.

A–B _____ C _____ D–E _____ Total _____

Where did pace change, and what decision caused it?

Confidence audit No return

List two items you would most want to revisit on a linear test. For each, state the verification you used **before** committing instead.

G Error Log

Up to 5 corrections

Choose one primary type: Meaning, Ratio Order, Whole, Scaling, Arithmetic, Changed State, Target, or Format.

Problem	First wrong decision	Primary type	Next-time interruption cue

H Transfer Reflection

Complete sentences

1. The label or model that prevented my most likely ratio error was …

Sample: Labeling the ratio first quantity : second quantity and using a total-parts model helped me avoid reversing the ratio or leaving a group out of the whole.

2. One clue that a question requires total parts rather than a simple scale factor is …

A clue is that the problem gives a combined total and asks for one group, so I must add all ratio parts before finding the value of one part.

3. When a ratio story changes one group, my first observable action will be …

I will label the original counts, apply the stated change only to the affected group, and rebuild the total before forming the new ratio.

4. My five-second commit question for a proportional relationship will be …

My question will be, "Did I preserve the required ratio or unit rate and answer the requested quantity in the required format?"